Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

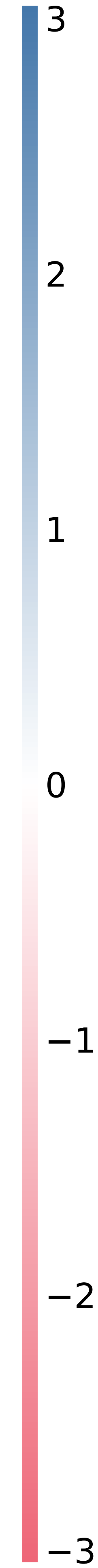
F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think> To find the current drawn by the headlight bulb, we need to use the formula: P = V ^ 2 / R, where P is the power, V is the voltage, and R is the resistance. However, we are asked for current, which is given by I = P / V. So we will use the given power (P = 40 watts) and voltage (V = 12 volts) to find the current. The first formula can be rearranged to I = sqrt(P/R). Then we use I = P / V to derive I = V / sqrt(P/V). Where V / sqrt(P/V) = 12 / sqrt(40/12). Then it further simplifies to (12 * 12) / sqrt(40) = 144 / sqrt(40) or (12 ^ 2)/(sqrt(4*10) Or it is also (12 ^ 2)/sqrt(4* sqrt(10)). By approximating, it becomes 12 ^ 2 / (2* sqrt(10) which then again becomes approximately 36/2 * 3.16 / The approximate answer simplifies to 3.1587. Rounding it to single decimal place gives us 3.1 amps but any of the answer choices approximated to 3.0. Given R = V ^ 2 / P. Next we are asked to calculate the resistance through previous P V and R values given to us. , i.e R = (12 ^ 2/40) then, simplifies R = 9/5 or simplifies to 1.8 which R and drawn is given as one of answers 3.0 a ms with an approx answer R = 3.0 which appears to be 4 sec oh ms given roughly html correspond class another valu with come through min ulle imaginary approx precedence registered manual Gui http </think><answer> A </answer>